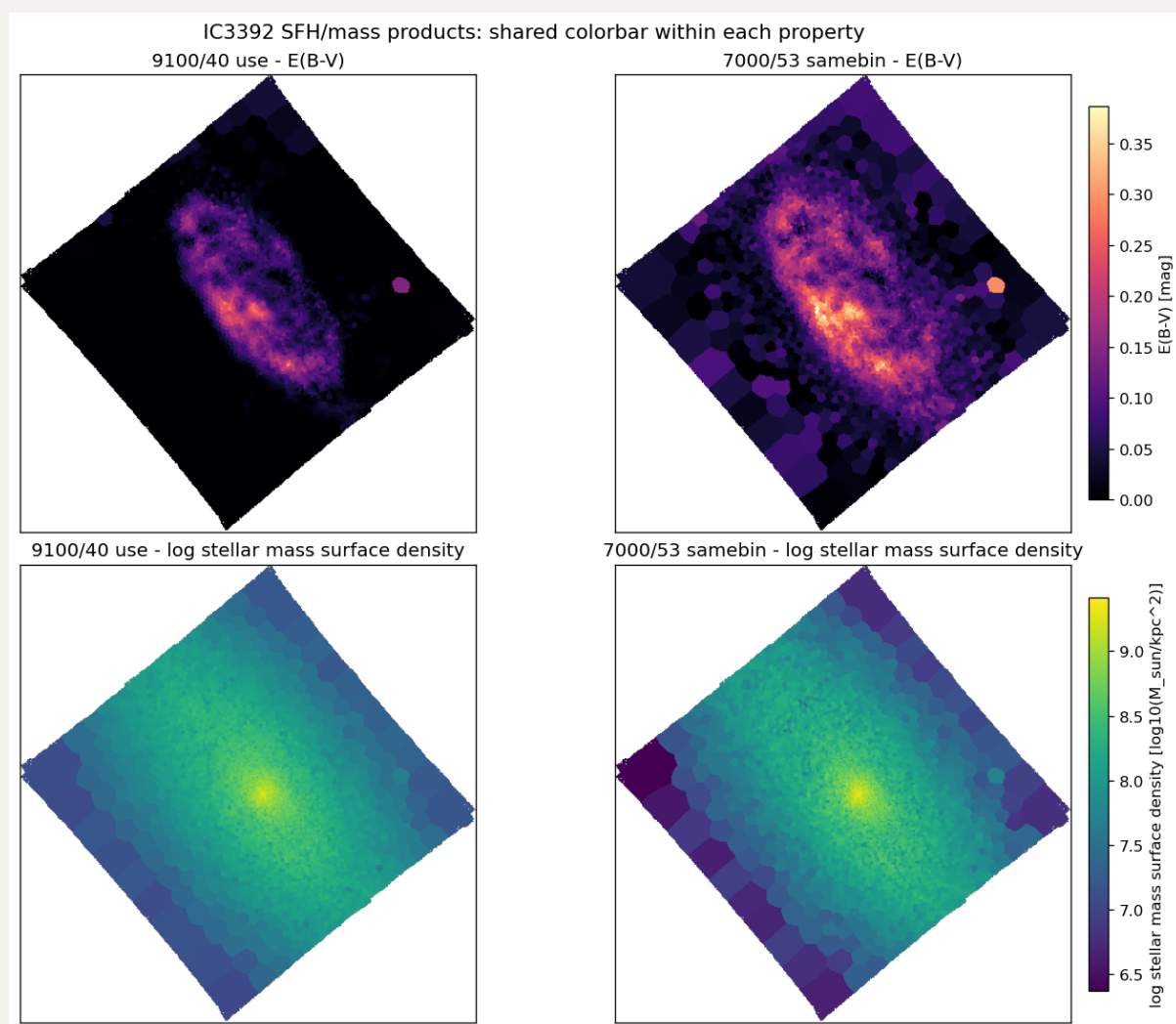
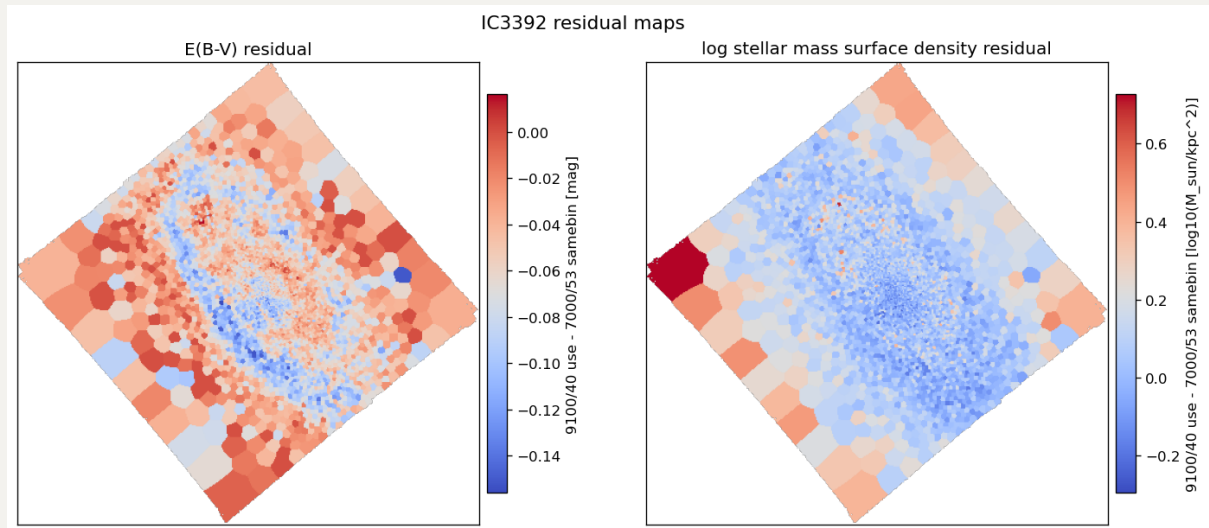
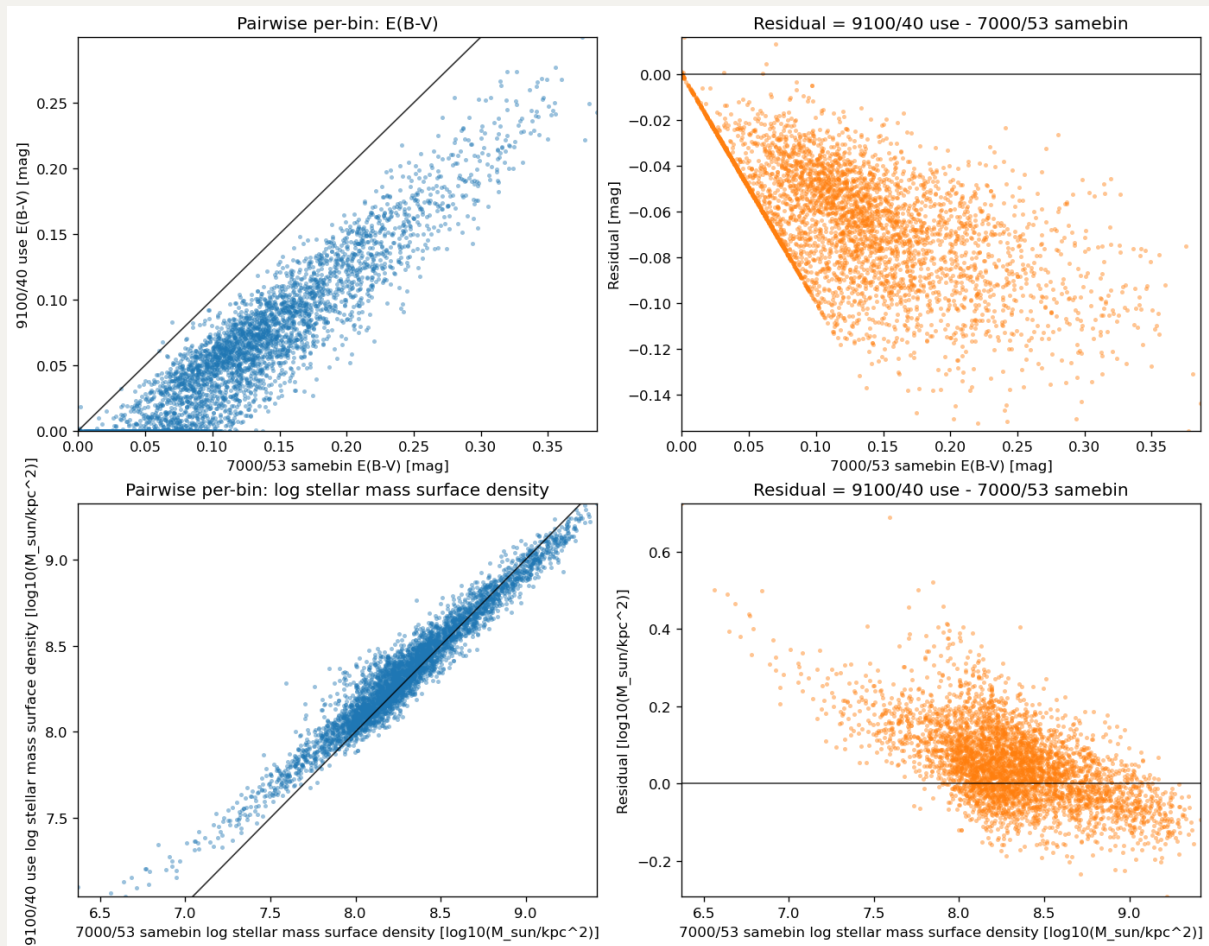


20260518 SFH and Stellar Mass Check

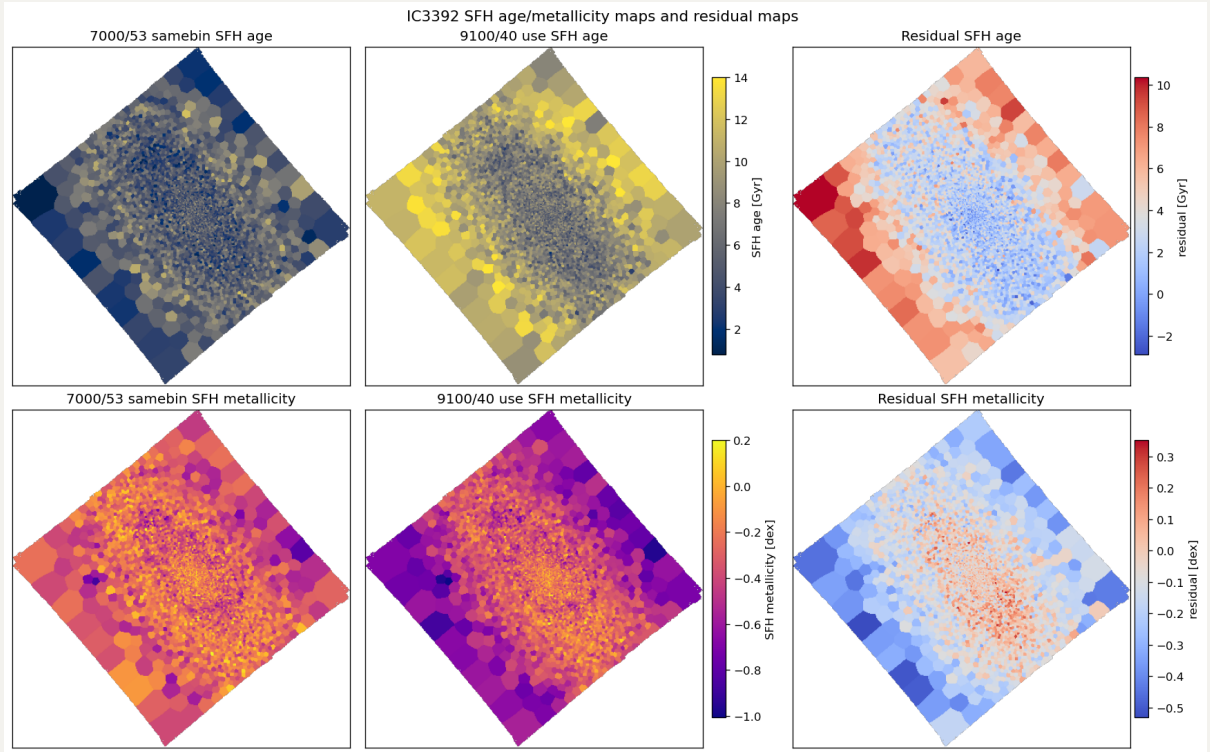
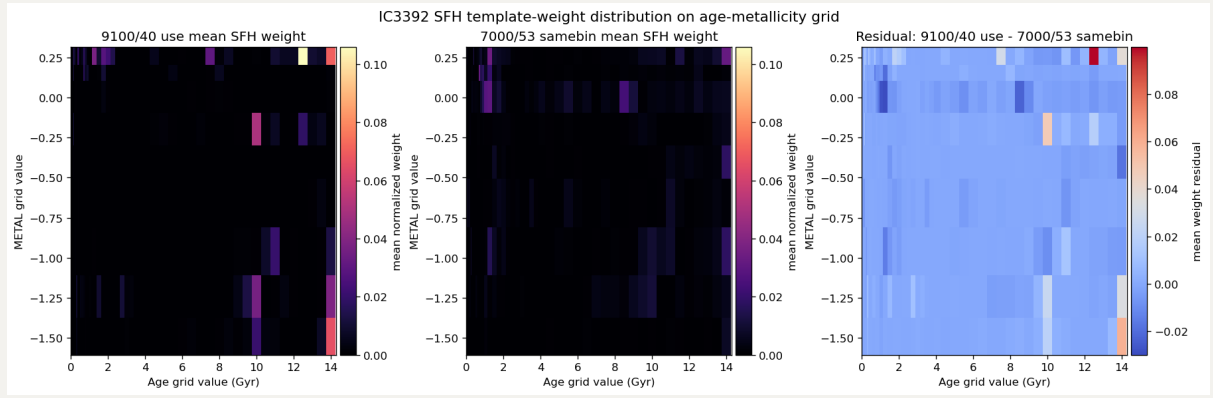
1. IC3392

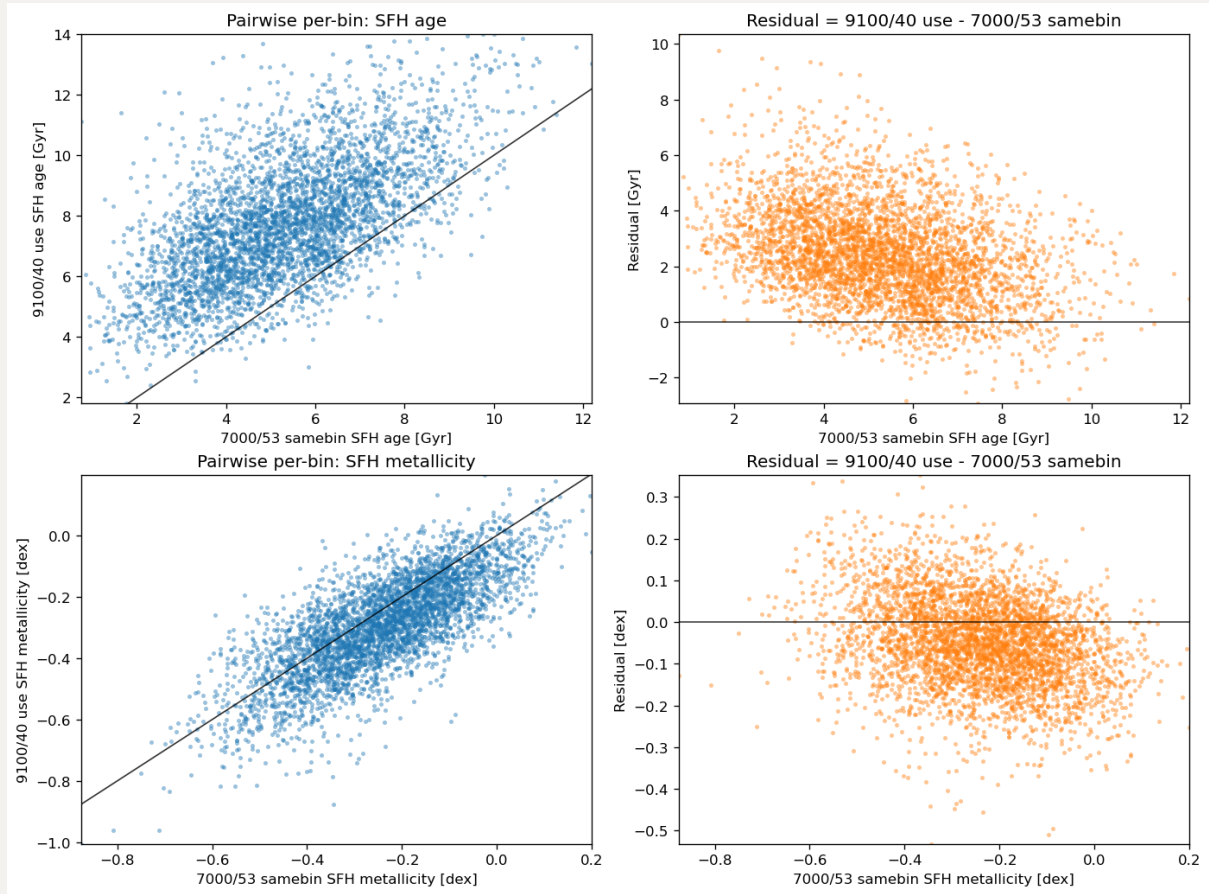
1.1 IC3392 E(B-V) and stellar mass maps.





1.2 IC3392 SFH weights: Age and stellar metallicity [M/H]





1.3 IC3392 Integrated values comparison

	QUANTITY	7000/53 SAMEBIN	9100/40 USE	9100/40 USE - 7000/53 SAMEBIN	UNIT
0	Total R-band magnitude (uncorrected)	12.2600	12.2600	0.000	mag AB
1	Total R-band magnitude (corrected)	11.8650	12.0770	0.212	mag AB
2	Total R-band luminosity	9.5330	9.4480	-0.085	log10(Lsun)
3	Total stellar mass (R)	9.7210	9.7790	0.058	log10(Msun)
4	Integrated stellar M/L_R	1.5402	2.1402	0.600	Msun/Lsun

1.4 IC3392 Sumary

For $E(B-V)$, the two maps have very similar spatial structure, but the 9100/40 use run is systematically lower than the 7000/53 samebin run: the median value shifts from 0.1168 mag in 7000/53 samebin to 0.0560 mag in 9100/40 use. Almost all common bins move downward: 3986 / 4032 bins have negative residuals, only 7 bins are positive, and 39 are exactly zero.

For `LOGMASS_SURFACE_DENSITY`, the median surface density changes from `8.2787 dex` in `7000/53 samebin` to `8.3433 dex` in `9100/40 use`.

The uncorrected total R-band magnitude is identical in the two logs, `12.260 AB`. After extinction correction, however, `9100/40 use` is fainter: `12.077 AB` compared with `11.865 AB`, giving a lower total R-band luminosity by `0.085 dex`.

The total stellar mass is higher in `9100/40 use` by `0.058 dex`, because the integrated `M/L_R` is much higher: `2.1402` compared with `1.5402 M_sun/L_sun`.

In SFH-weight-grid check, the `9100/40 use` run shifts toward older templates: the mean linear age from the weight grid increases from `5.321 Gyr` to `7.750 Gyr`, and the fraction of weight at ages `>= 8 Gyr` increases from `0.367` to `0.570`.

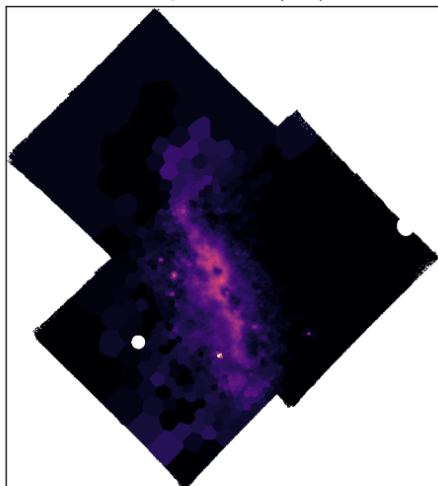
The spatial `AGE` and `METAL` maps show the same trend. The median fitted age increases from `5.242 Gyr` in `7000/53 samebin` to `7.638 Gyr` in `9100/40 use`, with a median residual of `+2.324 Gyr`; `3770 / 4032` bins are older in the `9100/40 use` run. The median metallicity decreases from `-0.238 dex` to `-0.286 dex`, with a median residual of `-0.0498 dex`; `2757 / 4032` bins are more metal-poor in the `9100/40 use` run. Age is therefore the stronger coherent shift, while metallicity also moves slightly lower.

2. NGC4383

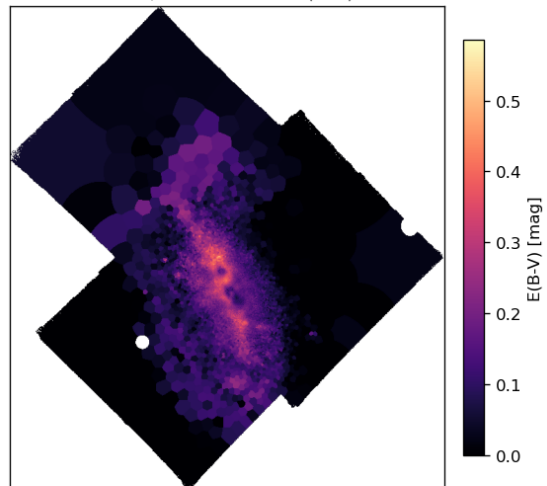
2.1 NGC4383 E(B-V) and stellar mass maps.

NGC4383 SFH/mass products: shared colorbar within each property

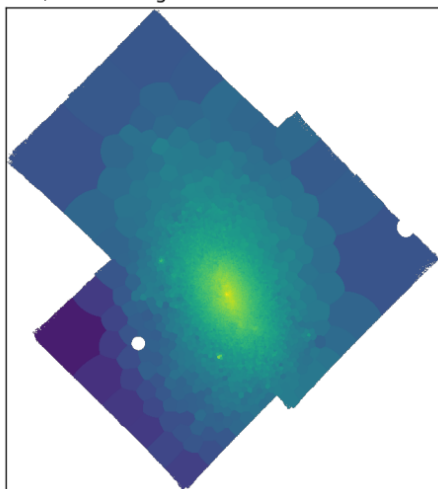
9100/40 use - E(B-V)



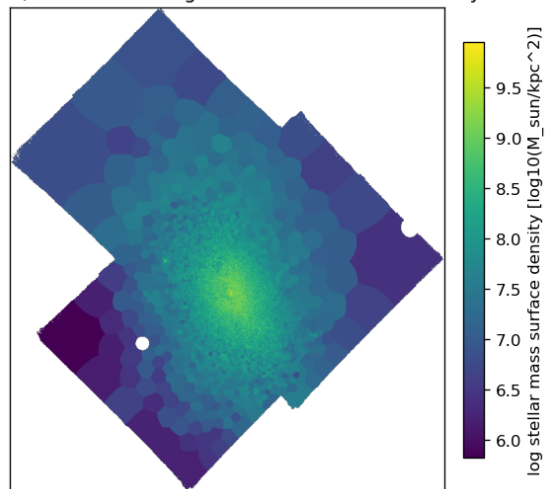
7000/53 samebin - E(B-V)

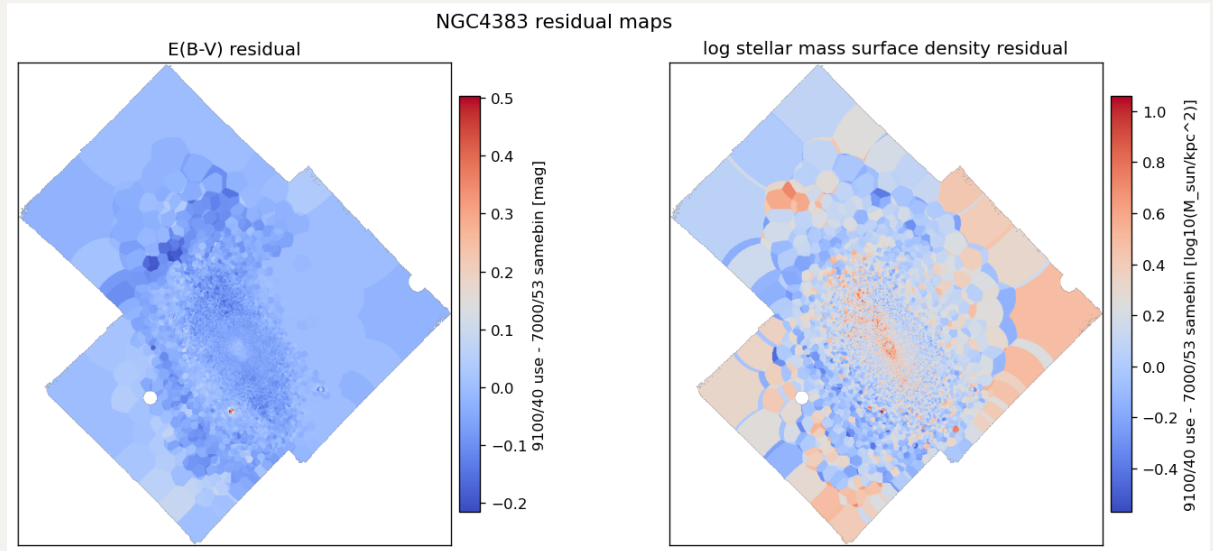
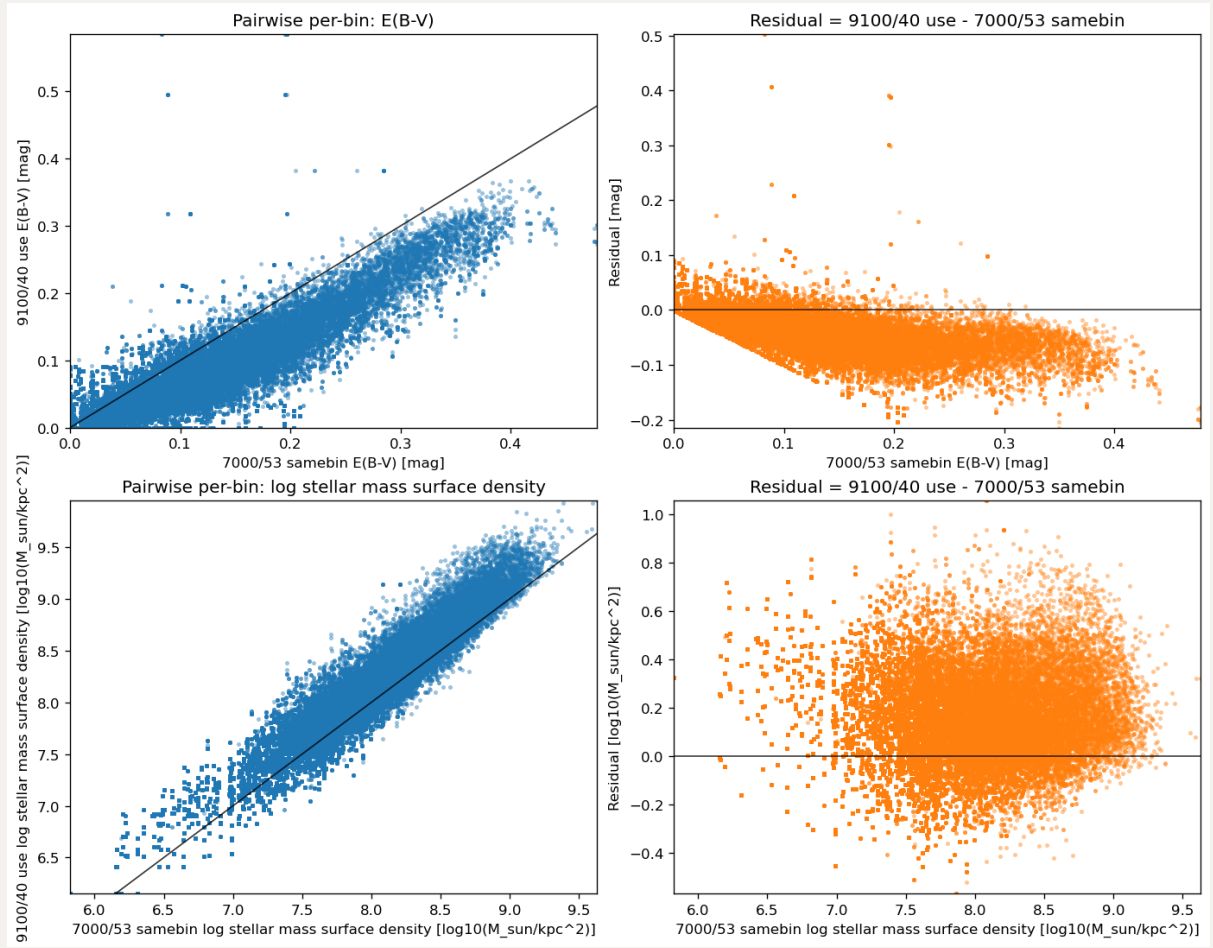


9100/40 use - log stellar mass surface density

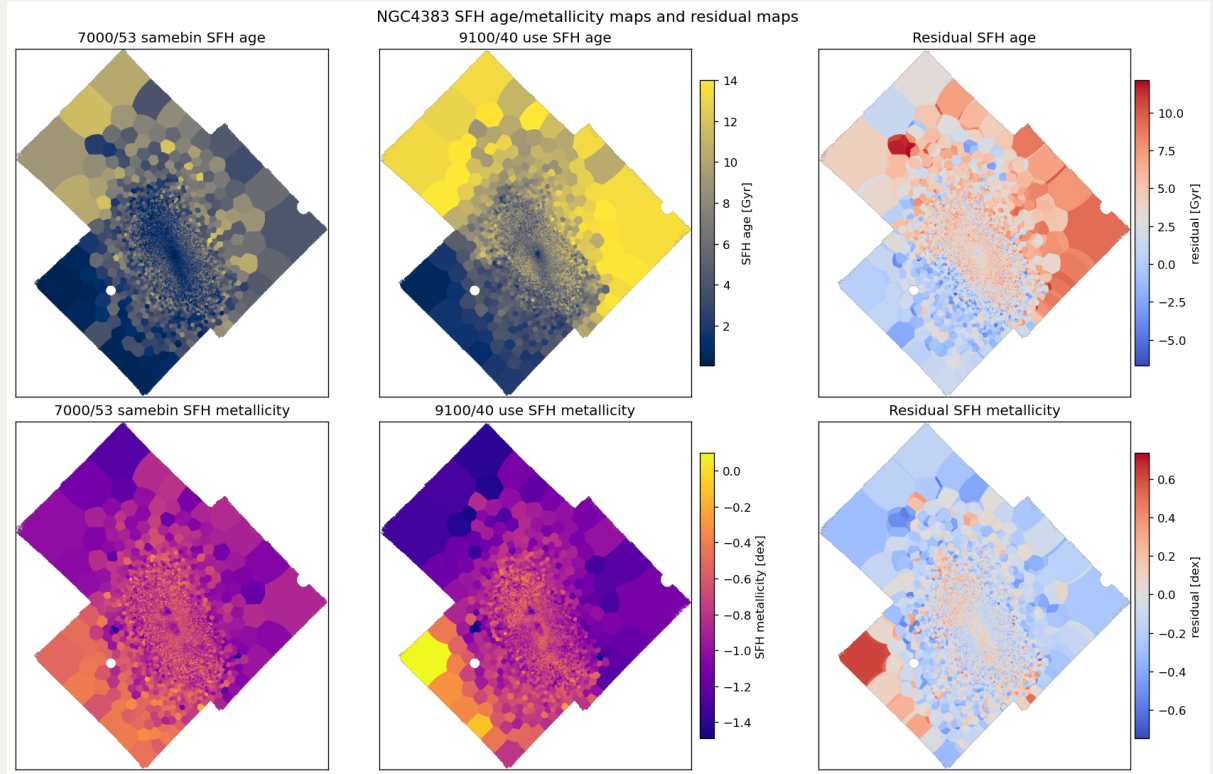
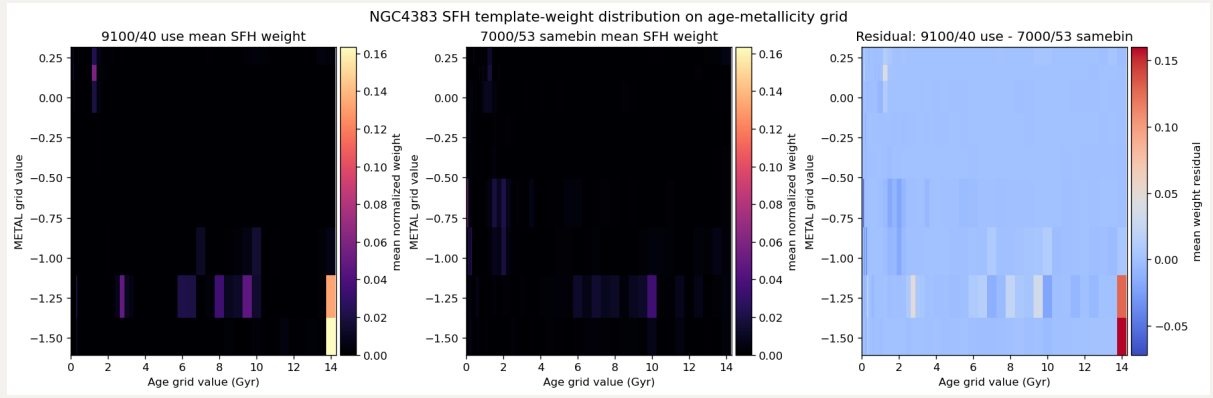


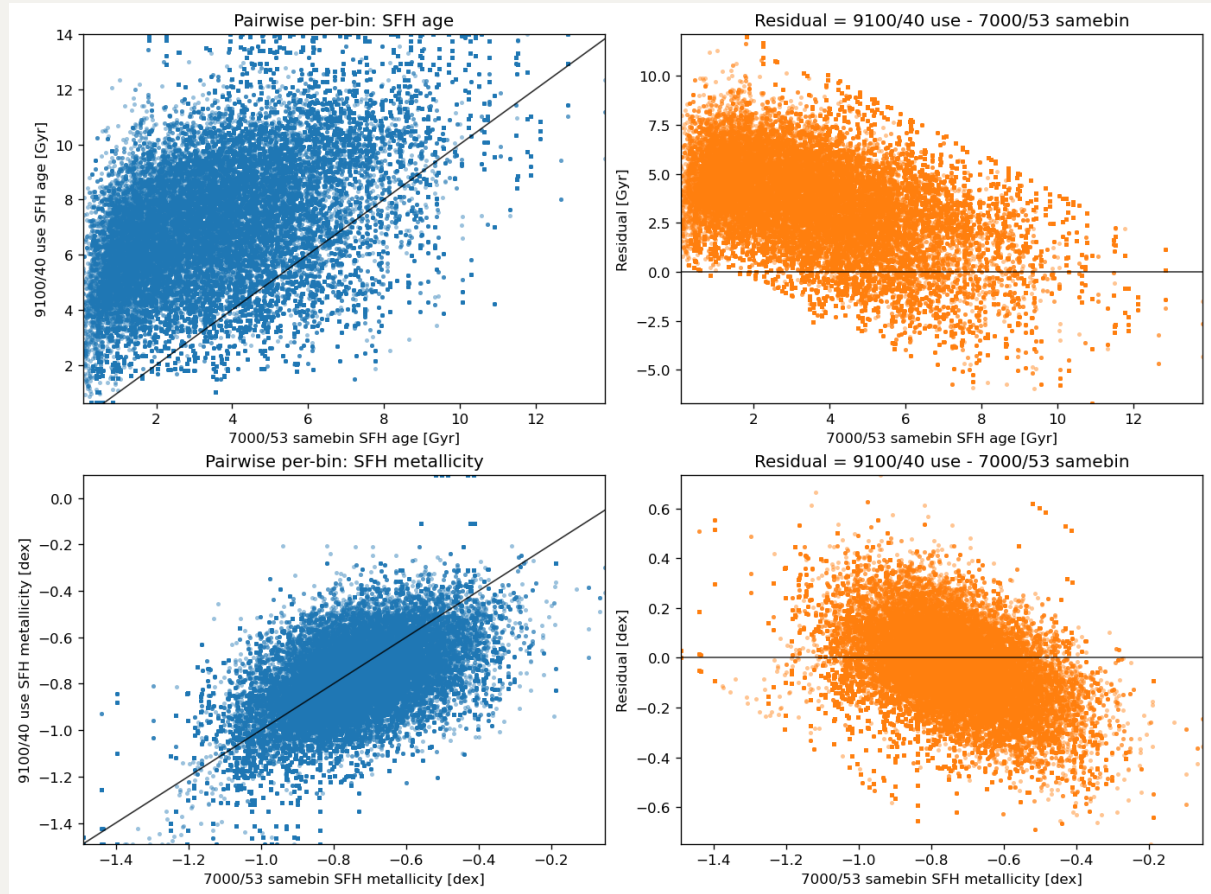
7000/53 samebin - log stellar mass surface density





2.2 NGC4383 SFH weights: Age and stellar metallicity [M/H]





2.3 NGC4383 Integrated values comparison

	QUANTITY	7000/53 SAMEBIN	9100/40 USE	9100/40 USE - 7000/53 SAMEBIN	UNIT
0	Total R-band magnitude (uncorrected)	11.8420	11.8420	0.0000	mag AB
1	Total R-band magnitude (corrected)	11.3020	11.4450	0.1430	mag AB
2	Total R-band luminosity	9.7580	9.7010	-0.0570	log10(Lsun)
3	Total stellar mass (R)	9.6200	9.7980	0.1780	log10(Msun)
4	Integrated stellar M/L_R	0.7279	1.2494	0.5215	Msun/Lsun

2.4 NGC4383 Sumary

For $E(B-V)$, the spatial pattern is broadly preserved but the 9100/40 use run is lower overall. The median value changes from 0.0249 mag in 7000/53 samebin to 0.0243 mag in 9100/40 use.

For $\text{LOGMASS_SURFACE_DENSITY}$, the median changes from 7.0749 dex to 7.2259 dex.

The parsed mass logs show that the uncorrected total R-band magnitude is unchanged at 11.842 AB. After extinction correction, 9100/40 use is fainter by 0.143 mag, corresponding to a total R-band luminosity change of -0.057 dex.

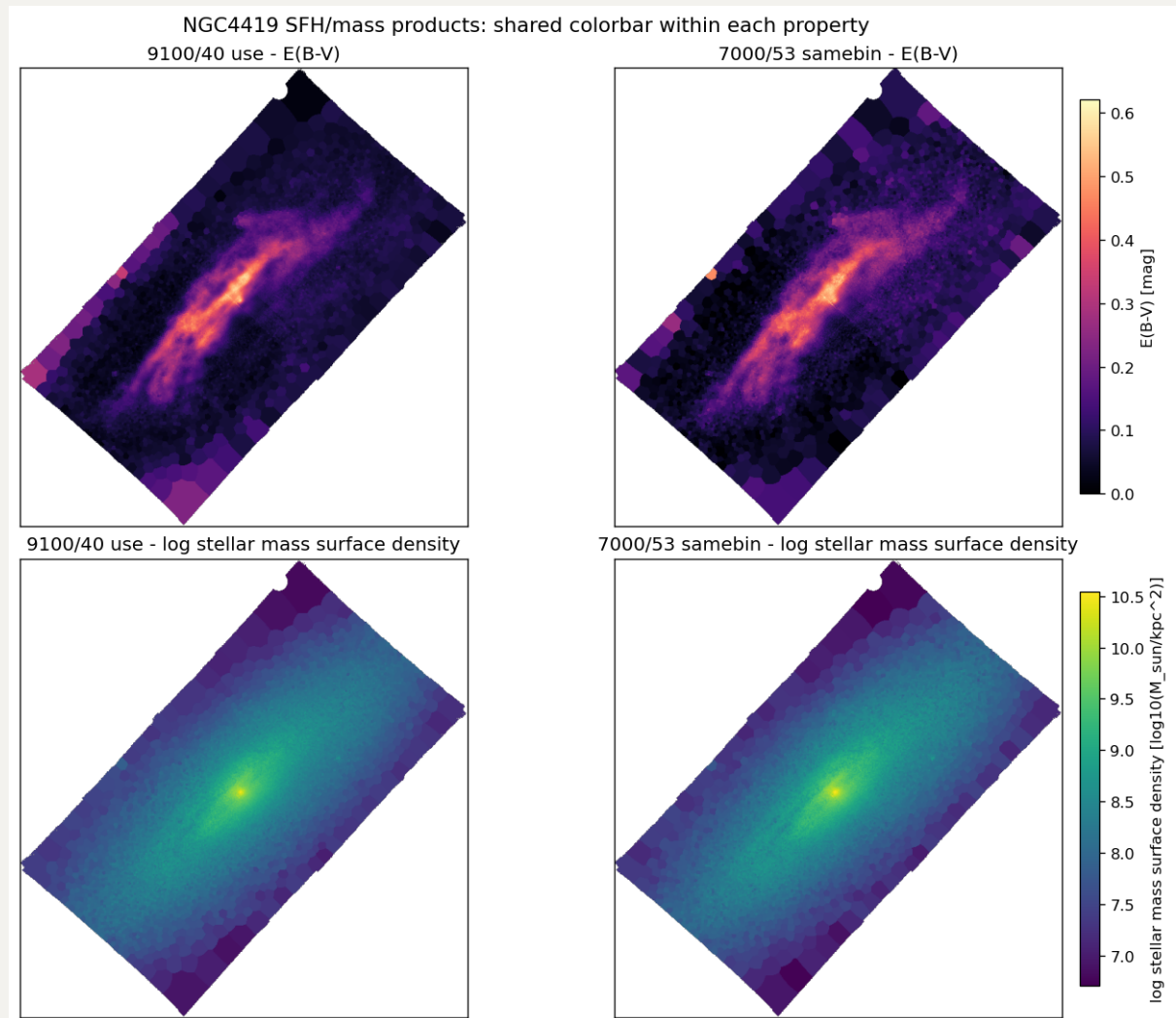
The total stellar mass changes by 0.178 dex, and the integrated M/L_R changes from 0.7279 to 1.2494 $M_{\text{sun}}/L_{\text{sun}}$.

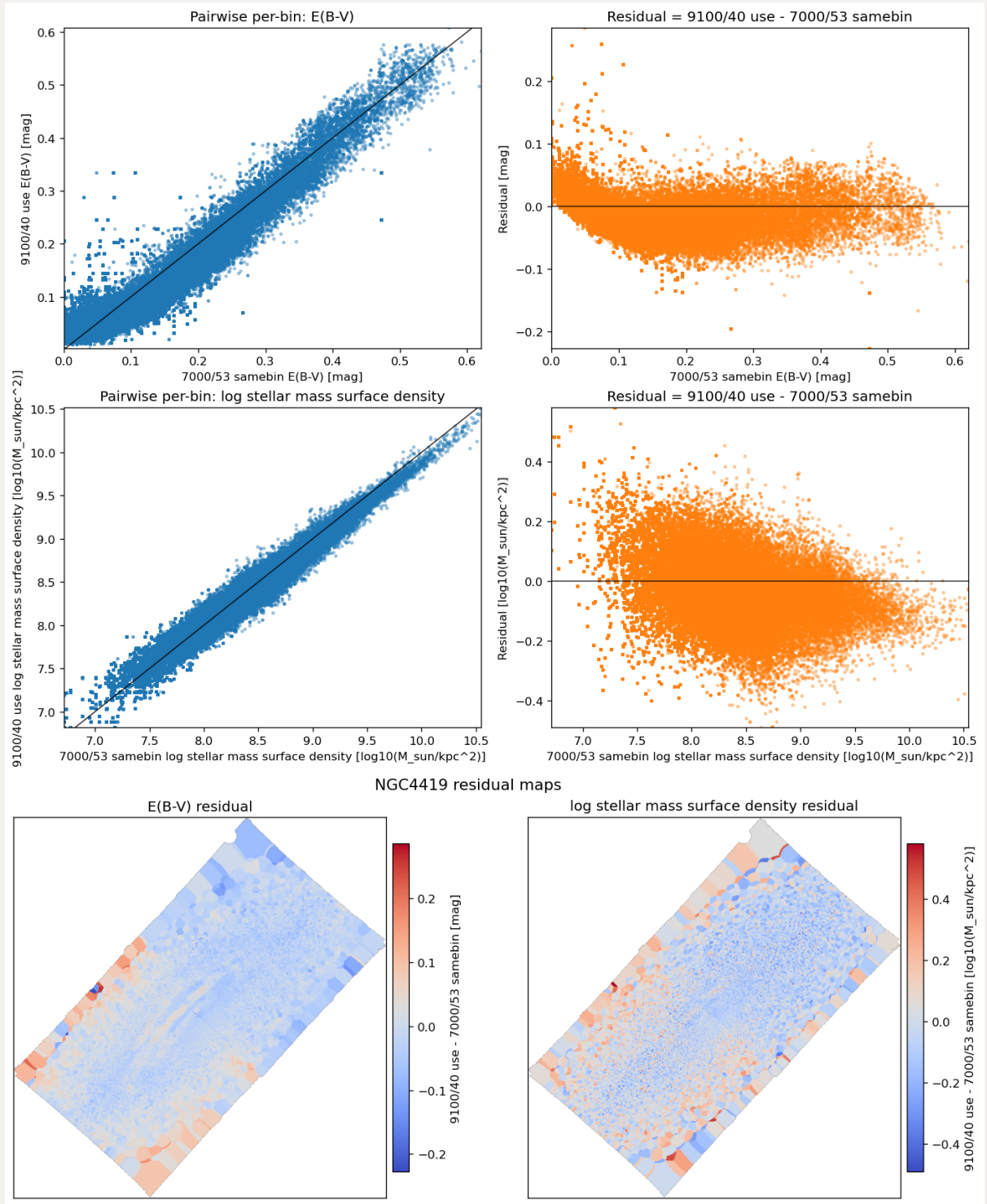
In SFH weight distribution, the mean age changes from 3.022 Gyr to 6.822 Gyr, and the weight fraction at ages ≥ 8 Gyr changes from 0.168 to 0.450.

The spatial AGE and METAL products provide the map-level view of the same SFH changes. The median age changes from 4.986 Gyr to 10.354 Gyr. The median metallicity changes from -0.8717 dex to -0.9836 dex.

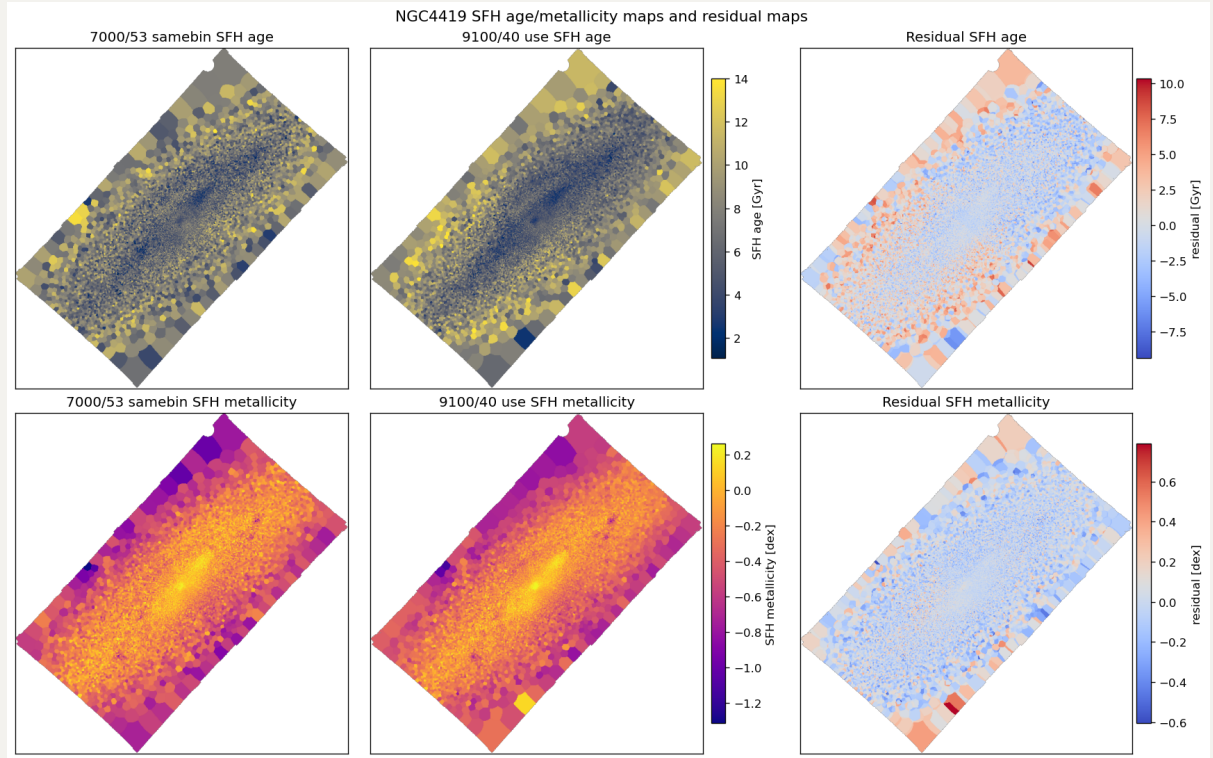
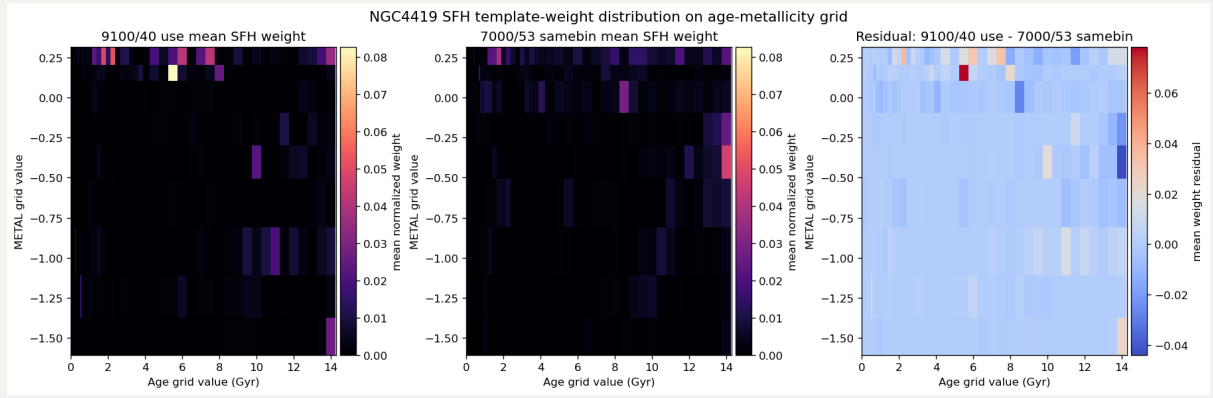
3. NGC4419

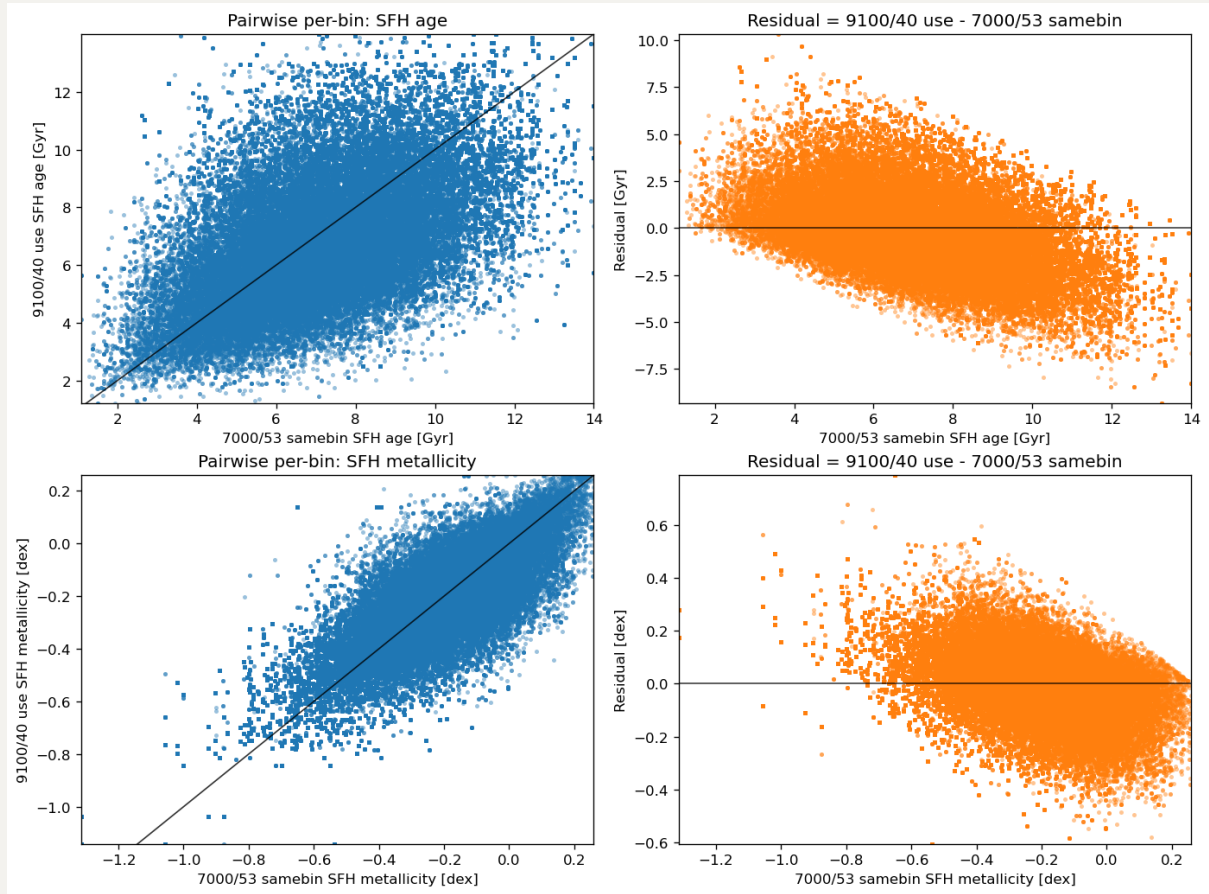
3.1 NGC4419 E(B-V) and stellar mass maps.





3.2 NGC4419 SFH weights: Age and stellar metallicity [M/H]





3.3 NGC4419 Integrated values comparison

	QUANTITY	7000/53 SAMEBIN	9100/40 USE	9100/40 USE - 7000/53 SAMEBIN	UNIT
0	Total R-band magnitude (uncorrected)	10.8590	10.8590	0.0000	mag AB
1	Total R-band magnitude (corrected)	10.1380	10.2030	0.0650	mag AB
2	Total R-band luminosity	10.2240	10.1980	-0.0260	log10(Lsun)
3	Total stellar mass (R)	10.5090	10.4540	-0.0550	log10(Msun)
4	Integrated stellar M/L_R	1.9295	1.8032	-0.1263	Msun/Lsun

3.4 NGC4419 Sumary

For $E(B-V)$, the spatial pattern is broadly preserved but the 9100/40 use run is lower overall. The median value changes from 0.0896 mag in 7000/53 samebin to 0.0754 mag in 9100/40 use.

For $\text{LOGMASS_SURFACE_DENSITY}$, the median changes from 7.8453 dex to 7.8497 dex.

The parsed mass logs show that the uncorrected total R-band magnitude is unchanged at 10.859 AB. After extinction correction, 9100/40 use is fainter by 0.065 mag, corresponding to a total R-band luminosity change of -0.026 dex. The total stellar mass changes by -0.055 dex, and the integrated M/L_R changes from 1.9295 to 1.8032 M_sun/L_sun.

In SFH weight grids: the mean age changes from 6.608 Gyr to 6.064 Gyr, and the weight fraction at ages ≥ 8 Gyr changes from 0.431 to 0.293.

The spatial AGE and METAL products provide the map-level view of the same SFH changes. The median age changes from 7.759 Gyr to 8.061 Gyr. The median metallicity changes from -0.3495 dex to -0.3435 dex.